Program : **Diploma in Instrumentation Engineering**	
Course Code : **5082**	Course Title: **Industrial Automation and Control**
Semester : **5**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4**	Periods per semester: **60**

Course Objectives:

- To understand the basic concepts of different control strategies.
- To describe the use of computers in process Industry
- To solve the automation related problems using PLC ladder logic programming
- To get the knowledge about various modern control strategies
- To conceptualize safety and documentation in process fields

Course Prerequisites:

Topic	Course code	Course Title	Semester
Feedback control and stability		Control engineering	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Describe various process control strategies	15	Understanding
CO2	Solve the engineering problems using PLC	14	Applying
CO3	Describe Modern Control Strategies	14	Understanding
CO4	Explain Safety and documentation in process industries	15	Understanding

	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	3	3					
CO3	2						
CO4	2				3		2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Describe various process control strategies**		
M1.01	Explain different types of process variables	3	Understanding
M1.02	Compare Feedback and Feed forward control system	4	Understanding
M1.03	Explain Cascade control system, Ratio control system and Adaptive control system	4	Understanding
M1.04	Explain Inferential control system, Split Range Control system, Override control system and Auctioneering control system	4	Understanding
Contents: Basic Process control strategies -Single variable - Independent variable - Interactive variable - compound variable - multivariable control (with examples) - Feedback control system - Feed forward control system – Comparison - Cascade control system - Ratio control system - Adaptive control system -Inferential control system - Split Range Control system - Override control system-Auctioneering control system.			
CO2	**Explain the use of Computers in Process Control and solve the engineering problems using PLC**		
M2.01	Explain Data Loggers , Data Acquisition System, Supervisory control and Direct Digital Control	4	Understanding
M2.02	Describe Distributed control system	2	Understanding
M2.03	Explain Programmable Logic Controller and its application	3	Understanding
M2.04	Solve engineering problems using PLC	2	Applying
M2.05	Explain Supervisory Control And Data Acquisition System	3	Understanding

	Series Test - I	1	`
Contents: Computers in process control-Data Loggers with block diagram - Data Acquisition System with Block Diagram - Supervisory control with block diagram - Direct Digital Control with block diagram – Comparison - DCS with its architecture - general features – advantages – PLC - basic concept - Block diagram of PLC operation - input Module, Output Module - PLC scan time - List the programming methods - ladder diagram programming - simple program - SCADA with block diagram - general features - system components			
CO3	**Describe Modern Control Strategies**		
M3.01	Explain Virtual instrumentation	2	Understanding
M3.02	Understand LabVIEW and its applications.	2	Understanding
M3.03	Understand the concept of Artificial Neural Network	3	Understanding
M3.04	Compare the various Learning methods used in ANN	2	Understanding
M3.05	Understand the concept of Fuzzy logic	1	Understanding
M3.06	Explain the Block Diagram of fuzzy controller	2	Understanding
M3.07	Understand Intelligent Control and its Features	2	Understanding
Contents: Modern Control Strategies: Virtual Instrumentation - compare Virtual instrument and traditional Instrument - Hardware and software - Test control and design - LabVIEW - Text based programming and graphical programming - Advantages of LabVIEW – ANN - Biological Neuron - Artificial Neuron Model - Types of activation functions - concepts of Learning methods (supervised, Unsupervised and reinforcement) -Fuzzy Logic - Concept - Classical and fuzzy sets- Properties - Membership function – example - Fuzzy Logic System - Block Diagram - Artificial Intelligence - Intelligent Control – Features			
CO4	**Explain Safety and documentation in process industry**		
M4.01	Understand the basic classification and selection of equipments based on hazardous area	3	Understanding
M4.02	Explain the concept of Instrument protection and types	3	Understanding
M4.03	Describe Fire and Gas system	3	Understanding
M4.04	Explain safety instrumentation system	3	Understanding
M4.05	Explain engineering documents used in Instrumentation	3	Understanding

	Series Test - II	1	

Contents:

Safety and documentation in process industry: Hazardous area classification - basic concept – classification - selection of equipment based on hazardous area - Instrument protection – Concept – types - Fire and Gas (F & G) system - concept and block diagram (types of fire and gas detectors) - Safety Instrumentation System (SIS) - concept and block diagram - different layers - Instrumentation documents - Process Flow Diagrams (PFDs), P&I Diagram -Instrument loop diagrams – Graphical element - line symbol - Instrument identification from P&I Diagram – Examples - Heat exchanger and Distillation column

Text / Reference:

T/R	Book title/Author
R1	Curtis .D.Johnson, “Process control instrumentation technology”, PHI
R2	George stephanopoulus, “Chemical Process Control”, PHI
R3	Krishnakanth, “Computer based Industrial Control”, PHI
R4	John webb, “Programmable Logic Controllers”, Macmillan Publishing
R5	Liptak,” Handbook of Process Control”
R6	Jovitha Jerome,”Virtual Instrumentation using LabVIEW”, PHI
R7	Rajasekaran. S. Vijayalakshmi Pai. G.A. “Neural Networks, Fuzzy Logic and Genetic Algorithms”, Prentice Hall of India Private Limited, 2003
R8	Timothy J.Ross, “Fuzzy logic with Engineering Applications”, McGraw Hill, 1995
R9	Alireza bahadori, “Hazardous area classification in petroleum and chemical plants a guide to mitigating risk”

Online Resources:

Sl.No	Website Link
1	https://www.slideshare.net/debuddit/hazardous-area-classification-56437172
2	https://youtu.be/IBJnU1MJAts